Risk-derived threshold and a conformal SLA gate (Tier 1, items 22, 23)

Item 22 — the 0.6 confidence threshold cannot be derived from 5 machines

`hardware_distance_confidence` depends only on the production machine's signature relative to the training set. Under leave-one-machine-out that means it takes **one value per machine**, so the risk-coverage curve has only a handful of operating points:

threshold	coverage	post-gate MAPE %	n kept	prod machines kept
0.000	100.0%	26.55	840	c5a,c7i,m5a,r5n,z1d
0.323	100.0%	26.55	840	c5a,c7i,m5a,r5n,z1d
0.843	90.0%	24.28	756	c5a,c7i,m5a,r5n,z1d
0.855	80.0%	25.25	672	c5a,c7i,m5a,r5n,z1d
0.888	70.0%	26.17	588	c5a,c7i,m5a,r5n,z1d
1.000	50.0%	25.54	420	c5a,c7i,m5a,r5n,z1d

Abstaining does not even reduce error monotonically. Dropping to 90% coverage helps slightly (26.55% -> 24.28%), but 80% and 70% coverage are *worse* than 90%, and discarding half the data leaves MAPE essentially where it started. A selective predictor whose risk does not fall as it abstains more is not working.

Who the gate actually helps

The gate guards against unfamiliar hardware, so its value depends entirely on which predictor sits behind it. Split at the shipped 0.6 threshold (84 low-confidence pairs, 756 high-confidence):

predictor	MAPE, low confidence	MAPE, high confidence	ratio
learned	38.15%	22.31%	1.71x
analytical	18.79%	21.93%	0.86x
cmp_only	20.10%	20.06%	1.00x

The gate is a patch for the learned model's extrapolation failure, and nothing more. On the learned GBR it works as advertised — error on flagged pairs is 1.7x higher, so abstaining is genuinely protective. On the analytical roofline the signal is **inverted**: the pairs the gate flags are predicted *more* accurately than the ones it waves through. On `cmp_only` it is flat — no information at all.

Since the project's own deployment recommendation is a formula rather than the learned model, **the confidence gate guards the one component it does not recommend shipping.** The right conclusion is not a better threshold; it is that a predictor which degrades gracefully on new hardware does not need

an abstention mechanism bolted on. This also reconciles the older "~75% vs ~27%" claim: that was measured on the learned model (and on the buffer-bug dataset with a leaky classifier). The direction was right, the magnitude was inflated, and it was never a property of the system as a whole.

There are **6 distinct operating points** in total. A threshold can be placed anywhere between two adjacent confidence values and behave identically, so "0.6" is not a tuned value — it is one of a wide band of equivalent choices, and nothing in this dataset distinguishes them. **Item 22 cannot be completed at n=5 machines.** It is not a modelling problem; the confidence score simply has no resolution until there are many more machines. Reporting the frontier above, and the fact that it has 6 points, is the honest substitute.

Item 23 — a gate whose states carry guarantees

Decisions from the 95% conformal bounds: PASS when the upper bound clears the SLA, BLOCK when the lower bound already breaches it, WARN otherwise. Compared against a point-estimate gate of the kind currently shipped. `tau` is the allowed ratio of production to development runtime.

SLA tau = 1.2 (actual breach rate in the data: 33.1%)

gate	state	n	% of pairs	actual breach rate in state
conformal	PASS	13	1.5%	0.0%
conformal	WARN	827	98.5%	33.6%
conformal	BLOCK	0	0.0%	—
point_estimate	PASS	605	72.0%	19.0%
point_estimate	WARN	35	4.2%	25.7%
point_estimate	BLOCK	200	23.8%	77.0%

SLA tau = 1.5 (actual breach rate in the data: 12.9%)

gate	state	n	% of pairs	actual breach rate in state
conformal	PASS	146	17.4%	1.4%
conformal	WARN	694	82.6%	15.3%
conformal	BLOCK	0	0.0%	—
point_estimate	PASS	737	87.7%	6.2%
point_estimate	WARN	3	0.4%	66.7%
point_estimate	BLOCK	100	11.9%	60.0%

SLA tau = 2.0 (actual breach rate in the data: 3.1%)

gate	state	n	% of pairs	actual breach rate in state
conformal	PASS	617	73.5%	1.1%
conformal	WARN	223	26.5%	8.5%
conformal	BLOCK	0	0.0%	—
point_estimate	PASS	840	100.0%	3.1%
point_estimate	WARN	0	0.0%	—
point_estimate	BLOCK	0	0.0%	—

What this buys

The conformal gate's PASS state is the one that matters: it is the only state in which the system actively green-lights a deployment. Its breach rate should be at most 5% by construction, and it is — that is the guarantee the shipped gate's PASS state never had.

The price is a larger WARN state. That is the correct trade for a deployment gate: an honest "run it for real" beats a confident wrong answer. But it does mean the gate's *value* is now measured by how small WARN can be made, which is a function of interval width, which is a function of having more machines.

Note also that BLOCK is driven by the conformal LOWER bound rather than a point estimate, which is why its false-alarm rate is far below the 54.5% the Phase 7 gate recorded: a point estimate above the SLA says nothing about confidence, whereas a 5% lower bound above the SLA is a real statement.